

Linear Kalman Filter for 2D Motion Tracking

Description

This project implements a linear Kalman Filter for estimating two-dimensional position and velocity. The system dynamics are linear, and the measurement model consists of direct position measurements observed in a rotated sensor frame. The project includes both simulation of truth and measurements and recursive state estimation using a discrete-time Kalman Filter.

System Model

The system equations are:

$$\dot{\underline{x}} = \mathbf{F}\underline{x} + \mathbf{B}\underline{u} + \underline{\nu}, \quad \underline{\nu} \sim \mathcal{N}(0, \mathbf{Q})$$

$$\underline{z} = \mathbf{H}\underline{x} + \underline{\eta}, \quad \underline{\eta} \sim \mathcal{N}(0, \mathbf{R})$$

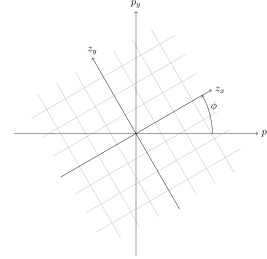
where

$$F = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.01 & 0 & 0 & 0 \\ 0 & 0.01 & 0 & 0 \\ 0 & 0 & 0.01 & 0 \\ 0 & 0 & 0 & 0.01 \end{bmatrix} \quad \underline{x} = \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix}$$

$$H = \begin{bmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \end{bmatrix} R = \begin{bmatrix} 1 & 0 \\ 0 & 0.1 \end{bmatrix} \underline{u} = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$$

and where (p_x, p_y) is the position, (v_x, v_y) is the velocity, (a_x, a_y) is the input acceleration, and θ is some arbitrarily chosen angle. There is a sensor at the origin of the coordinate frame directly measuring the position states. However, it is rotated by the angle θ . The $\mathbf{H}$ matrix represents this rotation from the state's coordinate frame to the sensor's coordinate frame:



Discretization

The continuous-time system is converted to a discrete-time model with sampling period $T = 0.1$ s. The discrete dynamics are written as

$$\underline{x}_{k+1} = \Phi \underline{x}_k + \mathbf{B}_d \underline{u}_k + \underline{\nu}_k, \quad \underline{\nu} \sim \mathcal{N}(0, \mathbf{Q}_d)$$

$$\underline{z}_k = \mathbf{H} \underline{x}_k + \mathbf{D} \underline{u}_k + \underline{\eta}_k, \quad \underline{\eta} \sim \mathcal{N}(0, \mathbf{R})$$

The matrices Φ , $\mathbf{B}_d$, and $\mathbf{Q}_d$ are computed using the van Loan method. This avoids relying on the first-order approximations and provides a more accurate discretization of both the dynamics and process noise.

Kalman Filter

State estimation is performed using a standard discrete-time Kalman Filter.

Update:

$$\mathbf{S} = \mathbf{H} \mathbf{P}_k \mathbf{H}^\top + \mathbf{R}$$

$$\mathbf{K} = \hat{\mathbf{P}} \mathbf{H}^\top \mathbf{S}^{-1}$$

$$\underline{r} = \underline{z}_k - \mathbf{H} \hat{\underline{x}}_k - \mathbf{D} \underline{u}_k$$

$$\hat{\underline{x}}_k = \hat{\underline{x}}_k + \mathbf{K} \underline{r}$$

$$\hat{\mathbf{P}}_k = \hat{\mathbf{P}}_k - \mathbf{K} \mathbf{H} \hat{\mathbf{P}}_k$$

Propagate:

$$\hat{\underline{x}}_{k+1} = \Phi \hat{\underline{x}}_k + \mathbf{B}_d \underline{u}_k$$

$$\hat{\mathbf{P}}_{k+1} = \Phi \hat{\mathbf{P}}_k \Phi^\top + \mathbf{Q}_d$$

The initial state estimate is taken from the first truth sample, and the initial covariance is chosen as the identity matrix.

Simulation setup

Truth data and measurements are generated over a 100 s simulation with sampling period 0.1 s. The acceleration inputs are defined as

$$a_x = -\sin\left(\frac{2\pi}{5}t\right) \quad a_y = -\cos\left(\frac{2\pi}{10}t\right)$$

The initial state is

$$\begin{bmatrix} 0 & 0 & 0.8 & 0 \end{bmatrix}^\top$$

Three cases are studied in the analysis:

1. Rotated measurement frame with $\theta = \pi/4$
2. Unrotated measurement frame with $\theta = 0$
3. Rotated measurement frame with isotropic measurement covariance $\mathbf{R} = \text{diag}[1.0, 1.0]$

Results

Figure 1 shows the true and estimated two-dimensional paths. The Kalman Filter tracks the state closely over the full simulation.

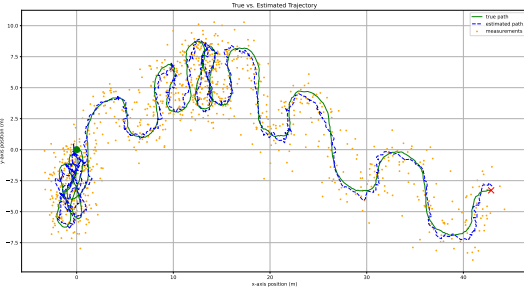


Figure 1: True and estimated 2D trajectories

Figures 2–5 shows the state estimation errors together with the expected $3 - \sigma$ bounds from the filter covariance and the empirical $3 - \sigma$ bounds computed from the full error history.

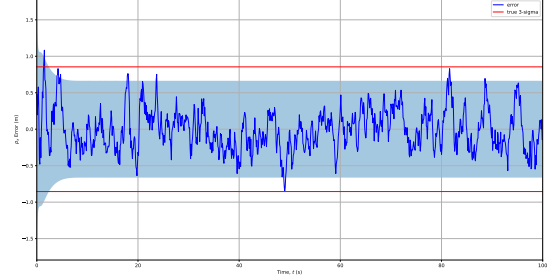


Figure 2: x-position error

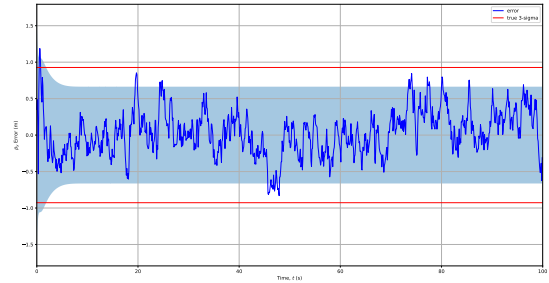


Figure 3: y-position error

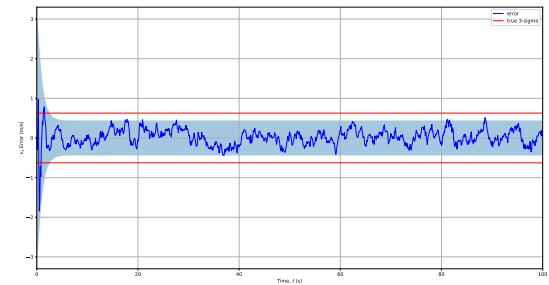


Figure 4: x-velocity error

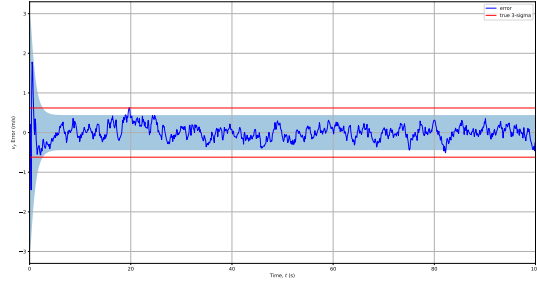


Figure 5: y-velocity error

Analysis

1. The final state covariance matrix of the estimation with rotated measurement frame with $\theta = \pi/4$

0.049	0.033	0.02	0.011
0.033	0.049	0.011	0.02
0.02	0.011	0.022	0.005
0.011	0.02	0.005	0.022

2. The final state covariance matrix after setting θ to 0.

0.082	0.0	0.03	0.0
0.0	0.016	0.0	0.009
0.03	0.0	0.027	0.0
0.0	0.009	0.0	0.017

There are 8 zeros in the new covariance matrix, as opposed to the matrix representing a $\pi/4$ radian rotation of the sensor. The presence of zeros in the aligned matrix is due to the fact that x-dimension and y-dimension become completely independent problems. When there is a rotation of the sensor, we must combine p_x and p_y in the H matrix to rotate our estimate into the sensor frame. This means that each measurement informs the filter about both p_x and p_y . When there is no rotation, we essentially get one measurement for p_x and one for p_y . When there is a rotation, the errors in the x and y states are *coupled*, meaning the errors in the x and y states are correlated. This is why there are no zeros in our original matrix, and zeros appear where x and y dimensions cross in the aligned matrix.

3. The final state covariance matrix after setting θ back to $\pi/4$ and setting R to $\text{diag}[1.0, 1.0]$.

0.082	0.0	0.03	0.0
0.0	0.082	0.0	0.03
0.03	0.0	0.027	0.0
0.0	0.03	0.0	0.027

There are once again 8 zeros in the covariance matrix, but this time appearing for a less intuitive reason (which is actually the dominant reason). The zeros reappeared because we restored perfect symmetry to the problem. The rotation of the sensor provides the mechanism to couple the x and y dimensions, but it is the *asymmetry* in the system that actually creates the correlation. When the measurement noise is asymmetric, the filter trusts one sensor direction more than another, forcing the x and y estimation errors to become correlated. By making the noise symmetric, any preference in direction is removed, so the filter treats both dimensions identically, causing the cross-covariance terms between them to resolve to zero.

Discussion

The results show that the Kalman Filter provides accurate state estimates while maintaining a covariance matrix that is consistent with the observed estimation errors. The comparison between the predicted and empirical 3- σ bounds provides a useful check on whether the filter uncertainty is well calibrated.

The covariance analysis also illustrates how the measurement model affects state coupling. When the measurement frame is rotated, the position states are mixed in the sensor model, which produces nonzero off-diagonal covariance terms. When the measurement frame is aligned with the state coordinates, additional structure appears in the covariance matrix. Changing the measurement covariance to an isotropic form further alters this structure by making the measurement uncertainty symmetric in the rotated frame.